

Tutorial 5 – IS Infrastructure: The Case of Netflix

Objective

Understand the evolution and components of IS infrastructure and analyse how Netflix uses cloud technology to manage costs, scalability, and governance.

Duration

1 – 1.5 hours

Activity Instructions

1. Read Article (15 minutes)

Students should read the article:

[Netflix on AWS](#) — This reading provides an overview of how Netflix migrated its infrastructure to the cloud, what components it relies on, and how it manages scalability, cost, and governance.

2. Group Discussion (30 minutes)

In your project group, discuss the following questions:

- a. What *infrastructure evolution stage* (L6 pp.8) does Netflix represent?
- b. Identify *key infrastructure components* (L6 pp.9) used by Netflix.
- c. What *management challenges* (L6 pp.16-17) has Netflix faced (e.g., platform transition, governance, investment)?

Hint: Think about the complexity of migrating to the cloud, microservices, and cost monitoring.

- d. How does Netflix optimize performance and cost through the *Total Cost of Ownership (TCO) model* (L6 pp.18)?

3. Individual Submission (Post-Class)

- a. Each student must individually submit answers to the questions in a **PDF file of no more than two pages**, named: T5_<full-name>.pdf(e.g., T5_xiaoming.pdf).
- b. Students must submit their work **before the next tutorial** to receive the **full score**.

- c. **Late submissions will receive only half the score. Submissions later than 2 weeks will not receive any score.**

Quick Reading Aid:

Background

Netflix, originally a DVD rental company, has evolved into the world's leading subscription-based streaming service. To support its rapid growth and global expansion, Netflix made a strategic decision to migrate all its infrastructure to Amazon Web Services (AWS). This move marked a complete transition from traditional on-premises data centers to a fully cloud-based environment. Operating entirely on AWS, Netflix delivers seamless streaming experiences to over 230 million members worldwide in more than 190 countries.

Overview

This article outlines how Netflix built its global streaming platform by leveraging cloud infrastructure provided by Amazon Web Services (AWS). It describes the company's shift from on-premises data centers to a fully cloud-native architecture, highlighting the technical tools used to support streaming, scalability, and user experience.

Key technologies mentioned in the article include:

- **Amazon EC2** – for elastic computing and handling millions of simultaneous streams
- **Amazon S3** – for storing and delivering large volumes of video content
- **Amazon CloudFront** – for content delivery across global edge locations
- **Amazon EMR and Redshift** – for data analytics and operational intelligence
- **Amazon SageMaker** – for powering machine learning and personalization features
- **AWS Lambda** – for event-driven automation at scale

The article also touches on how these technologies enabled Netflix to operate more flexibly, deploy infrastructure across regions rapidly, and support continuous innovation in its services.